

Sajeevan (Saj) Veeriah

ROBOTICS, MECHATRONICS, AI/ML AND END-TO-END AUTOMATION ENGINEER

Geelong, VIC · +61 498 586 654 · sajeevanveeriah@gmail.com
[linkedin.com/in/sajeevan-veeriah](https://www.linkedin.com/in/sajeevan-veeriah) · sajeevanveeriah.github.io · Member, Engineers Australia

SUMMARY

Robotics and mechatronics engineer who builds complete systems across mechanical design, electronics, embedded firmware, controls, autonomy, software, data and validation, taking work from requirements and prototyping through commissioning, verification and handover. Experience spans ROS 2 mobile robotics and SLAM, embedded and IoT platforms, GMP smart-factory automation and autonomous mobile robot operations, automotive emissions and EV testing, and advanced manufacturing quality. Works the full engineering loop: sense, estimate, control, actuate, verify.

CORE SKILLS

Robotics and autonomy	ROS 2 Humble, Nav2, MoveIt 2, SLAM, A* and path planning, Gazebo, RViz, tf2, URDF, DDS, EKF and Kalman state estimation, sensor fusion, LiDAR, IMU, wheel odometry, kinematics, PID motion control, MiR autonomous mobile robots
Embedded and electronics	C, C++, ESP32, STM32, FreeRTOS, bare-metal firmware, UART, I2C, SPI, CAN, ADC, PWM, DMA, interrupts, PCB design, Altium, KiCad, board bring-up, signal conditioning, oscilloscopes, logic analysers, sensors, motors, drives, actuators
Controls and modelling	MATLAB, Simulink, PID tuning, feedback and cascade control, state estimation, Kalman and EKF, system identification, motion control, process control loops, VFDs, loop testing
AI, ML and data	Python, scikit-learn, OpenCV, YOLO, Pandas, NumPy, computer vision, anomaly detection, predictive maintenance, time series analytics, data pipelines, InfluxDB, Grafana
Software, Linux and DevOps	Python, C, C++, TypeScript, JavaScript, Git, GitHub, Linux, Docker, CI/CD, GitHub Actions, REST APIs, PostgreSQL, Supabase, Next.js, Vercel, Node-RED
Mechanical and product	SolidWorks, Fusion 360, 3D CAD, GD&T, mechanism design, prototyping, 3D printing, engineering drawings, instrumentation, design for manufacture, manufacturing support
Automation, industrial networking and IoT	Siemens TIA Portal, WinCC, PCS 7, iFIX, PVI+, IEC 61131-3 (LD, FBD, ST, SFC), HMI and SCADA, Modbus, PROFINET, OPC UA, MQTT, LoRaWAN AU915, ChirpStack, MikroTik, edge gateways, field devices
Validation, quality and delivery	FAT, SAT, commissioning, qualification, GAMP 5, ADR and EURO emissions testing, EV and PHEV range testing, CANoe, CANalyzer, ITPs, MDRs, material traceability, FMEA, root cause analysis, OH&S, ISO, IEC and AS/NZS standards, JIRA, Agile, Lean Six Sigma, KAIZEN
Wider engineering domains	Systems engineering, mechanical design and thermofluids, electrical power, IoT and edge-to-cloud telemetry, automotive validation, biomedical devices, regulated manufacturing, civil and structural awareness, aerospace, marine, rail, defence, mining, agriculture and energy, safety and cyber-physical security, project delivery and commissioning. Evidence tier for each domain is published at sajeevanveeriah.github.io/atlas .

EXPERIENCE

Automation and Controls Engineer <i>JAG Process Solutions Pty Ltd, Brunswick, VIC</i>	Jan 2026 to Jun 2026
- Delivered end-to-end automation and systems integration for pharmaceutical, biotech and food manufacturing under GMP, spanning control logic, HMI and process visualisation, field devices, drives, production data, testing and handover. - Executed an iFIX to PVI+ SCADA migration, converting application content item by item and verifying functional behaviour against the validated system, then produced FAT, SAT, commissioning, qualification and handover evidence. - Supported MiR autonomous mobile robot operations and material-flow workflows in a smart-factory environment, diagnosing faults across robot, control, network and production interfaces.	
Technical Officer, QA and OH&S <i>ABMARC, South Geelong, VIC</i>	Jul 2024 to Aug 2025
- Conducted ADR and EURO emissions, EV and PHEV range and vehicle systems testing using calibrated instrumentation, data acquisition and CAN tools, analysing deviations and separating instrument fault from vehicle behaviour and procedure error. - Prepared QA, regulatory, OH&S and test documentation for certification and audit, holding traceable configurations, results and technical findings, and maintained emissions test equipment for accuracy between programs.	

Consultant Engineer, IoT and Projects Administrator

Feb 2024 to Aug 2024

DuxTel Pty Ltd, Greater Geelong, VIC

- Designed and deployed end-to-end IoT telemetry linking sensors, embedded devices, gateways, Linux services, data pipelines and dashboards using ESP32, LoRaWAN AU915, MQTT, ChirpStack, InfluxDB and Grafana.
- Integrated CAN capture, GPS, a custom PCB, condition sensing and MikroTik networking for agricultural equipment monitoring, running field trials and supporting provisioning, fault finding and handover.

Production Line Worker, Team Lead, Cellar Hand, Technical and Maintenance Support

Aug 2022 to Feb 2024

IDL, Geelong, VIC

- Led a production line team across shift output, changeovers and handover while holding batch quality, hygiene and traceability discipline in a regulated beverage production environment.
- Provided technical and maintenance support to line and cellar equipment, carrying out first-level fault finding and corrective work to restore production and reduce downtime.

Production Line Operator, Quality Control, Development and Commissioning Support

Sep 2021 to Dec 2021

Carbon Revolution, Geelong, VIC

- Operated carbon fibre composite wheel production equipment and performed in-process quality control, recording measurement and traceability data against process specification.
- Supported development and commissioning activity on production equipment, assisting with trials, setup and verification of process changes in an advanced composites manufacturing line.

Undergraduate Quality Engineer

Mar 2018 to Mar 2020

Thornton Engineering Australia, Geelong, VIC

- Supported structural steel engineering and QA through drawing review, inspection and test plans, manufacturing data reports, inspection planning and material traceability.
- Maintained AS/NZS compliance records and traceable quality evidence across fabrication projects, and supported first-level engineering queries between design, workshop and inspection.

SELECTED PROJECTS

Autonomous Navigation Rover on ROS 2. Built a complete ROS 2 Humble autonomy stack on a differential-drive platform with LiDAR SLAM, Nav2 and A* planning, PID motion control and EKF fusion of wheel odometry and IMU data, validated in Gazebo and RViz for repeatable obstacle-aware navigation.

ESP32 Clinical Ataxia Assessment Device. Designed a clinically trialled measurement device on ESP32-WROOM with four Hall-effect sensors, 100 Hz acquisition, a real-time state machine, custom PCB and enclosure, Bluetooth connection to a companion phone app for live recording and display, and clinician reporting with CSV and PDF export. Validated against reference instruments in MATLAB across accuracy, direction, reversal, drift and temperature protocols.

Engineering Mastery Lab. Built and published a browser-first engineering workbench combining input-validated calculators, bounded parametric CAD, eight guided learning labs and evidence-focused project workflows across web and desktop modes, owning architecture from engineering logic and CAD geometry to security boundaries and validation.

EDUCATION

Bachelor of Mechatronics Engineering (Honours), Distinction

2025

Deakin University, Geelong, VIC

Higher National Diploma, Mechatronics, Robotics and Automation Engineering, Distinction

2016

Cardiff Metropolitan University, United Kingdom

ADDITIONAL DETAILS

Membership and licence	Member, Engineers Australia. Current Victorian driver licence.
Community	Secretary and digital coordinator, Newcomb and District Cricket Club. Built and maintain the club production website on Next.js, TypeScript, Supabase, Vercel and Stripe.
Training and languages	Lean Six Sigma Foundation, JIRA and Agile, KAIZEN, Industrial Automation and IIoT, AI/ML, CAD. English, Tamil, Sinhala.

REFERENCES

Professional and community references available on request.